Dalton's Law of Partial Pressures:-

Consider a vessel with a mixture of two or more gases

↓

Individual Pr.s of the different gases will depend upon the relative proportions of the constituents

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Individual Pr.s are usually expressed as the partial Pr.s of the constituents, defined as the Pr. it would exert if it alone occupied the entire volume of the vessel at the same Temp.

↓

Dalton studied this and stated as: The total Pr. of a mixture of gases is equal to the sum of the partial Pr.s of the constituent gases at the same Temp.

↓

If a mixture of gases (at temp. T) contains n_a, n_b, n_c, …. moles of different gases whose partial Pr.s are P_a, P_b, P_c,…. etc. respectively, the total Pr. P of the mixture is;

$$P = P_a + P_b + P_c + \cdots$$

Since, $P = \frac{n}{V} RT$

$$P_a = \frac{n_a}{V} RT, \quad P_b = \frac{n_b}{V} RT, \quad P_c = \frac{n_c}{V} RT$$

Hence, $P_a + P_b + P_c + \cdots = (n_a + n_b + n_c + \cdots) \frac{RT}{V}$

$$= \frac{n}{V} RT$$

Therefore, $P = P_A + P_B + P_C + \cdots$

Also, $P_A = \frac{n_a}{n} P, \quad P_B = \frac{n_b}{n} P, \quad P_C = \frac{n_c}{n} P, \ldots$

Law of Partial Volumes (Amagat):-

The total vol. of a gas mixture is the sum of the partial volumes of the constituents of the mixture.

Thus, $V = v_1 + v_2 + v_3 + \cdots$

V = Total vol. ; $v_1, v_2, v_3 \cdots$ are the partial volumes of the constituents.

$$v_1 = \frac{n_1}{P} RT \; ; \quad v_2 = \frac{n_2}{P} RT, \; \cdots$$

$$V = v_1 + v_2 + v_3 + \cdots = (n_1 + n_2 + n_3 + \cdots) \frac{RT}{P}$$

$$= \frac{nRT}{P}$$

Moreover, $v_1 = \frac{n_1}{n} V, \quad v_2 = \frac{n_2}{n} V, \; \cdots$

Ideal Gases

$PV = nRT$ → The gas which obeys this equn. under all conditions of temp. and Pr. is called ideal gas. The gas which does not obey this equn. at all temp.s and Pr.s is called Real gas.

Ideal gas:-

① The ideal gas cannot be liquefied. As the gas has no intermolecular attraction, so the molecules will not be condensed.

② co-efficient of thermal expansion (α) depends solely on Temp (T) and does not depend on the nature of the gas.

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$$

For 1 mole ideal gas, $PV = RT$, Hence, $\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P}$

So, $\alpha = \frac{1}{V} \times \frac{R}{P} = \frac{R}{RT} = \frac{1}{T}$

i.e, $\alpha = \frac{1}{T}$

Thus, all gases have the same co-efficient of thermal expansion at a given temp.

(3) Co-efficient of compressibility (β) is defined as

$$\beta = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$

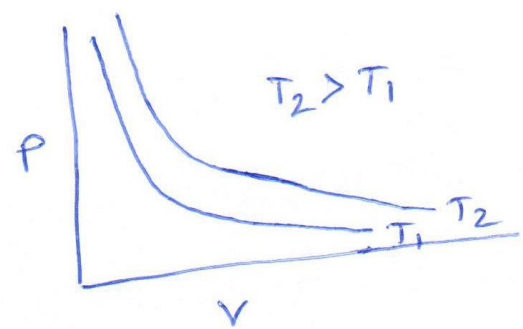
For ideal gas, $PV = RT$; So, $\left(\frac{\partial V}{\partial P}\right)_T = -\frac{RT}{P^2}$

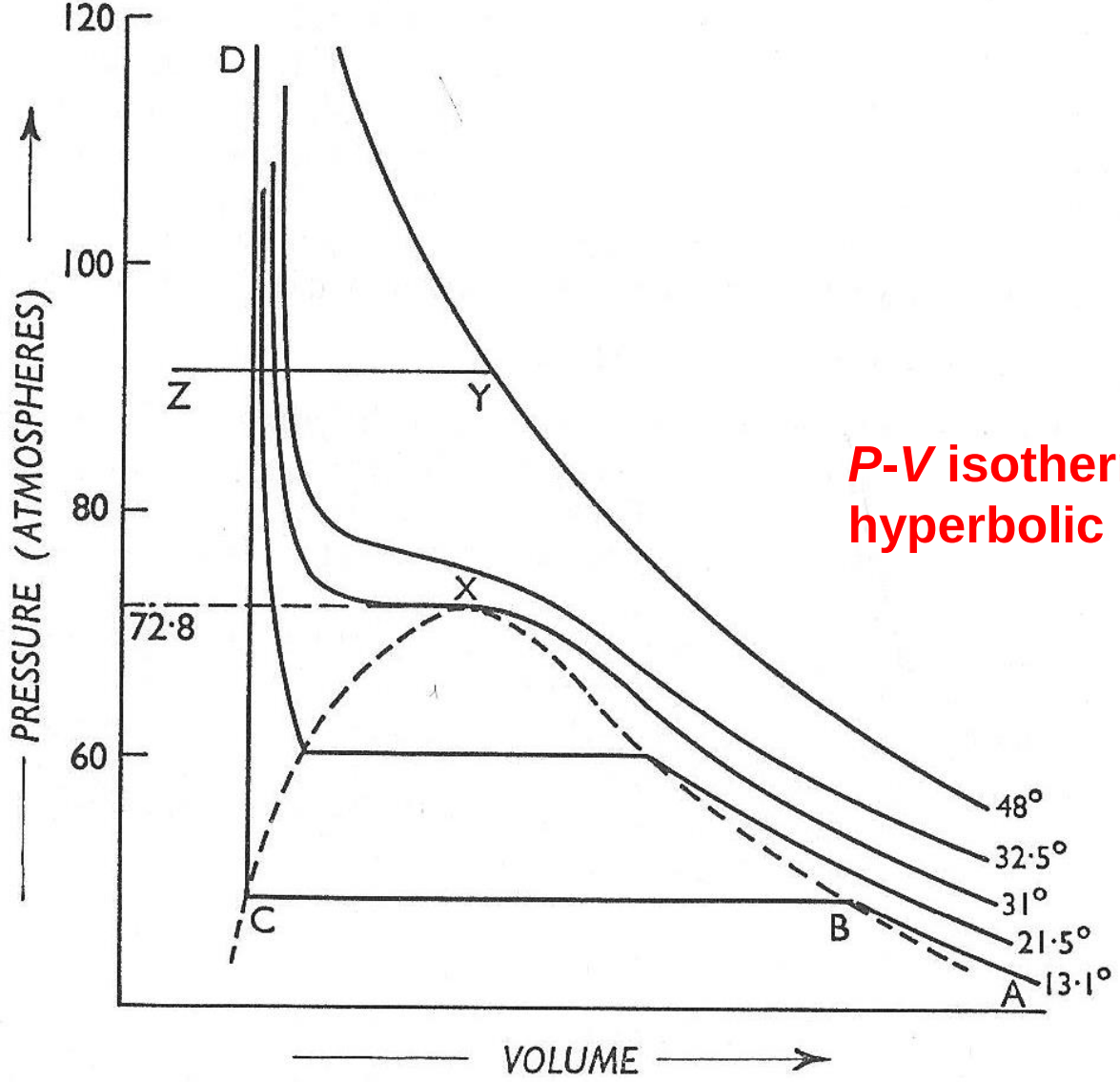
Thus, $\beta = -\frac{1}{V} \times -\frac{RT}{P^2} = \frac{RT}{PV \times P} = \frac{PV}{PV} \times \frac{1}{P} = \frac{1}{P}$

$\Rightarrow \beta = \frac{1}{P}$

Thus, β also depends only on P of the gas and same for all gases.

(4) when P is plotted against V at const. temp, a rectangular hyperbola curve is obtained as demonstrated by Boyles law, PV = const. at a given T. The hyperbola curve at each T is called one isotherm and at diff. temp.s we have diff. isotherms. Two isotherm will never intersect.

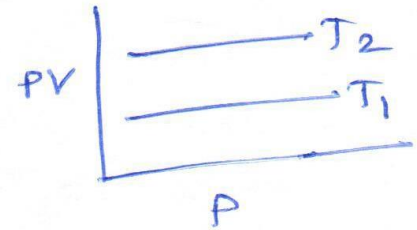




P-V* isothermals are found to be hyperbolic only at fairly high *T

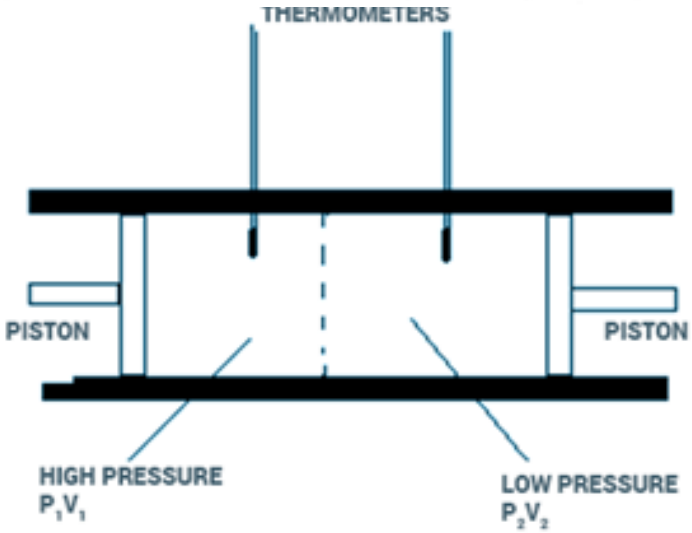
P *vs.* V curves for carbon dioxide

(5) when PV is plotted against P at const. T, a st. line parallel to P-axis is obtained.
At diff. temp.s there will be diff. parallel lines.



(6) when an ideal gas passes through a porous plug, from higher Pr. to lower Pr. under insulated enclosure, there will be no change of temp. of the gas (J-T expansion). This confirms that ideal gas has no intermolecular attraction.

JOULE - THOMSON EFFECT



Real gas :

① This gas could be liquefied since it has intermolecular attraction which helps to coalesce the gas molecules.

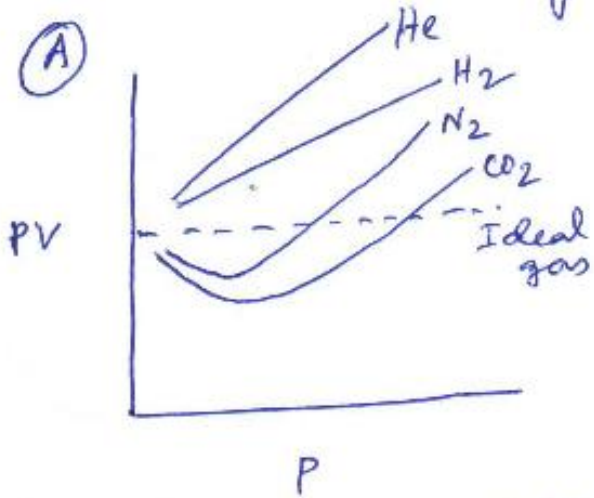
② The co-efficient of thermal expansion (α) is found to vary from gas to gas at a given temp. and hence it depends on the nature of the gas.

③ The co-efficient of compressibility (β) also is found to depend on the nature of the gas at a given Pr.

④ when P is plotted against V at a const. temp, a rectangular hyperbola is obtained only at high T (above a certain T, called critical temp, T_c of the gas). T_c is defined as the temp above which the gas could not be liquefied whatever the Pr. is applied. But at temp. below T_c, the gas is liquefied after certain Pr. depending on Temp.

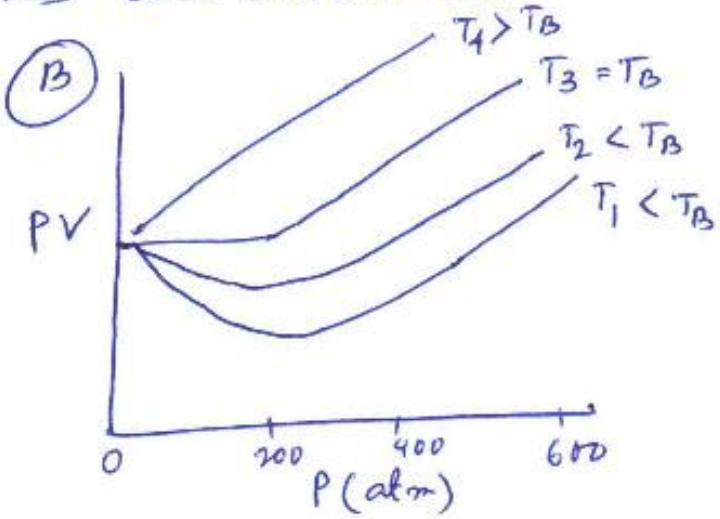
(5) when PV is plotted against P for real gas, the following plots; called Amagat's curves are obtained

(A)



Amagat curves for different gases at 0°C.

(B)



Amagat curve for CO_2 at diff. temp.s.

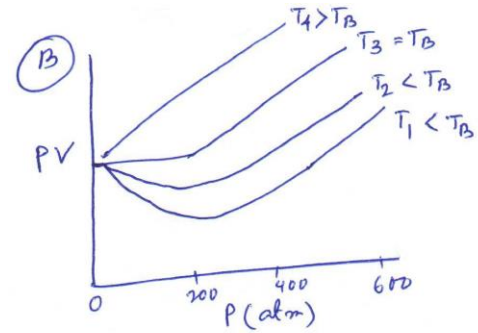
Fig A → shows for most gases, the value of PV decreases, attains minimum and then increases with increase of pr. Only H_2 and He baffle this trend and the curve rises with increase of P from the beginning.

Real Gases

Fig B → shows that for CO_2 gas, the depth of the minimum shifts towards the PV axis with increas of T. At T_3 temp., PV curve runs parallel to P-axis up to certain range of P at low pr. region ($P \to 0$). This temp. is called Boyle temp (T_B) at which real gas also obeys Boyle's law up to certain range of Pr. at the low Pr. region. The minimum coincides with the PV axis. The mathematical condition for calculation of Boyle temp (T_B) is:

$$\left[\frac{\partial (PV)}{\partial P}\right]_T = 0 \quad \text{when } P \to 0$$

$T_1 < T_2 < T_3 = T_B < T_4$



Amagat curve for CO_2 at diff. temp.s.

- ✓ **The curves obtained for H_2 and He at 0 °C is above their T_B**
- ✓ **So, with increase of P, the value of PV increases from beginning**

Compressibility factor (Z) is used to measure the extent of deviation of the real gases from ideal behaviour

$$Z = \frac{PV}{RT} \; ; \; V = \text{Molar vol. of the gas}$$

Z = 1 ; Ideal gas, there is no deviation of the gas from the ideal behavior

Z ≠ 1 ; Non-ideal gas, departure of the value of Z from unity is a measure of the extent of non-ideality of the gas

- ✓ ***T_c*** **→ The highest temperature at which gaseous state starts condensing into liquid state**
- ✓ ***T_B*** **→ The temperature at which real gases obey ideal gas equation over an appreciable pressure range**

Real Gases

$$Z = \frac{PV}{RT}$$

when $Z < 1$, the gas is more compressible

when $Z > 1$, the gas is less "

Since V is a function of T and P, Z is also a function of T and P, so Z may be defined as

$$Z = \frac{V}{V_{ideal}} \quad \text{and} \quad Z = \frac{P}{P_{ideal}}$$

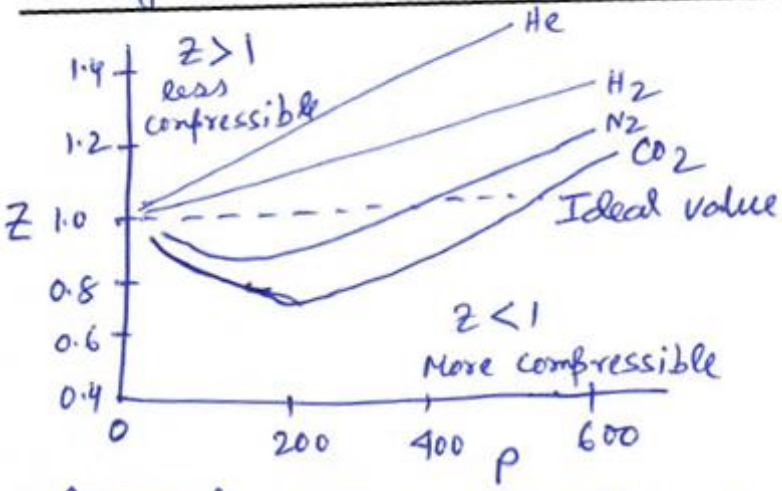
where, V_{ideal} is the molar volume of an ideal gas at the same T and P as the real gas.

Similarly, P_{ideal} is the Pr. of an ideal gas at the same T and V as the real gas.

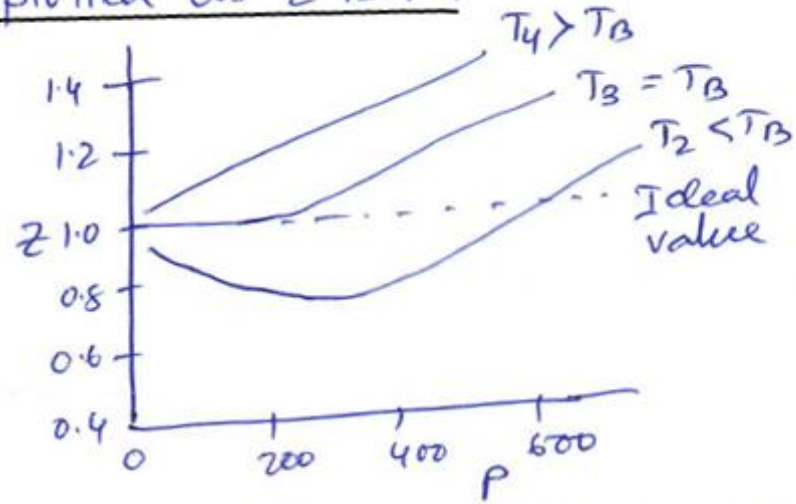
When $Z < 1$, the gas exerts lower Pr. than the ideal gas would and the volume of the gas becomes also lower than that of the ideal gas i.e. the gas becomes more compressible.

When $Z > 1$, the gas exerts higher Pr. than the ideal gas would and the volume of the gas becomes higher than that of the ideal gas i.e the gas becomes less compressible.

Amagat curves can also be plotted as Z vs. P:



Amagat curves for different gases at a given temp.



Amagat curves for a gas (CO_2) at diff. temperatures

T_B values:	**N_2 : 50 °C**	**CO_2 : 450 °C**
	H_2 : –164 °C	**CH_4 : 498 K**
	He : –250 °C	**O_2 : 406 K**

(6) When real gases pass through porous plug from higher pr. to lower pr. within insulated enclosure, there occurs a change of Temp. This is due to the fact that real gases have intermolecular attraction and when the gas expands, the molecules have to spend energy to overcome intermolecular attraction and so the temp. of the gas drops down.